

Crystal Characterization Study of Candidate Crystals for Calorimetry

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Abstract

High-resolution calorimetry for future collider experiments can be achieved by employing dual readout (DR) of Cherenkov and scintillation light. The CalVision consortium aims to realize this state-of-the-art calorimetric measurement for both homogeneous scintillating crystals and fiber-based hadronic sections [1], with a phased program that includes investigating candidate crystal materials for the electromagnetic calorimeter. This project aims to characterize the optical and scintillation behavior of Bismuth Germanate (BGO) crystal blocks to assess their performance as a potential candidate for electromagnetic calorimetry. The work involves performing (1) transmission measurements and systematic effects, and (2) light yield measurements to determine the scintillation response and decay characteristics using a custom setup and comparing the results with those obtained from the first project.

Introduction and Motivation

The signal produced by a calorimeter shower initiated by an incident hadron is generally lower than that of an electron/photon of the same energy, due to invisible energy losses from, e.g., nuclear binding energies, neutrinos, and particles with small inelastic cross sections such as neutrons escaping the detector or depositing energy outside of the sampling time window. The amount of missing energy is correlated with the number of inelastic nuclear collisions. Maximal-information calorimetry identifies measurable quantities correlated with the missing energy to enable shower-by-shower corrections. An example is dual-readout calorimetry, which uses scintillation light to measure the total ionizing energy deposit, and timing and/or wavelength filters to isolate and collect the Cherenkov light, which is produced only by relativistic particles in the shower (mainly electrons produced in pion showers).

As shown in Figure 1, the two types of light can be separated either by their wavelength (left), their arrival time (right), or polarization (not shown). As illustrated in the figure, the scintillation spectrum for PbWO_4 begins very close to where the transmission cuts off for small wavelengths, and the optimal region to select a Cherenkov signal is then above $\simeq 600$ nm. For scintillators such as Bismuth Germanate ($\text{Bi}_4\text{Ge}_3\text{O}_{12}$, commonly known as BGO), where the scintillation spectrum is further shifted from the transmission cutoff, a notch filter may be used to select light on both sides of the scintillation spectrum [2]. The peak is situated at 480nm as can be seen from Figure 2 and with their excellent transparency for wavelengths 330nm and over. Additionally, at room temperatures, it exhibits a singular dominant decay component with a time constant of 300ns, which is much larger compared to 15ns of PbWO_4 [3]. The longer decay constant allows better time resolution between the Cherenkov and scintillation signals

BGO is a high-density scintillating crystal widely used in radiation detection and calorimetry. It possesses several desirable properties: it is non-hygroscopic and mechanically robust, making it easy to handle; it is also relatively cost-effective compared to other inorganic scintillators. With a high density of 7.13 g/cm^3 and an effective atomic number of 74, BGO provides excellent γ -ray absorption efficiency [4]. Furthermore, its optical transparency and radiation hardness make it an attractive material for electromagnetic calorimeters, where a detailed understanding of its transmission and scintillation response is essential for achieving uniform light collection and precise energy measurement.

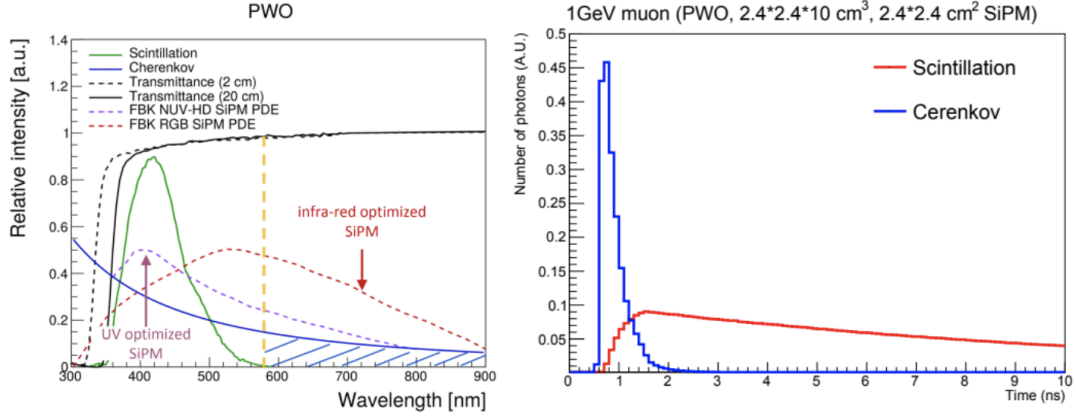


Figure 1: [left] Wavelength spectra of Cherenkov and scintillation emissions of $PbWO_4$ crystals along with the transmittance curves and photon detection efficiencies of example SiPMs. Hashed areas represent potential Cherenkov detection regions using wavelength filters. [right] The timing distributions of scintillation and Cherenkov light at the end of a 100 mm long $PbWO_4$ crystal with a cross-sectional area of $24 \times 24 \text{ mm}^2$, produced by 1 GeV muons crossing the middle of the crystal bar.[2]

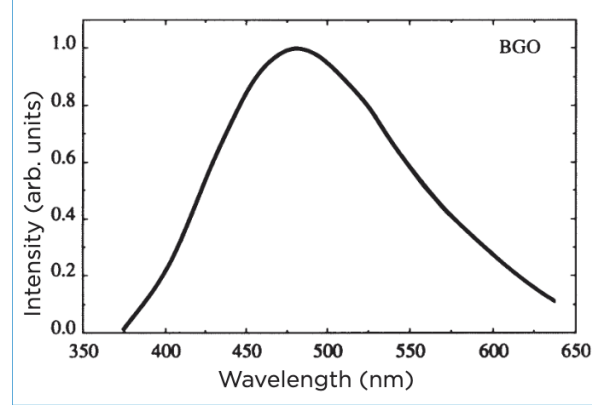


Figure 2: Scintillation emission spectrum of BGO [3]

2 Objectives

This project focuses on testing BGO crystal blocks ($22 \times 22 \times 160 \text{ mm}^3$) received from Shanghai Institute of Ceramics, Chinese Academy of Sciences(SICCAS) to evaluate their optical transmission and to measure their light yield and decay time using a controlled experimental setup. [5]. The proposed studies for this project involve

Transmission Properties: The goal is to quantify and compare the light transmission through BGO crystals using the spectrophotometer available at Prof. Tanja Horn's Nuclear Physics Laboratory at the Catholic University of America (CUA), with the assistance of graduate students Mr. Avnish Singh and Mr. Sameer Jain. Additional measurements will be done to help assess the *systematic uncertainties* affecting transmission, particularly those arising from misalignment of the crystal sample with respect to the emission axis by measuring the transmission spectrum for angular variations from -5° to $+5^\circ$ for both longitudinal and transverse orientations.

Light Yield Measurements: Parallel efforts are being carried out at the High Energy Physics (HEP) Laboratory at CUA and the HEP Laboratory at the University of Maryland (UMD), under the joint supervision of Prof. Shin-Shan Eiko Yu and Prof. Sarah Eno, to establish a radiation-based light yield measurement setup. The study involves selecting suitable photomultiplier tubes (PMTs), radiation sources, and developing a functional data acquisition (DAQ) system. The

choice of radiation source and PMT is important to ensure maximum light output given the relatively low light output of BGO (approximately 15–20% of NaI(Tl)) as well as the shifted emission spectrum of BGO (>500 nm) where phototubes are less sensitive [3]. We also aim to quantify and reduce the uncertainties arising from alignment and optical coupling between the crystal and the PMT. The resulting light yield measurements will be used to study the overall light output, longitudinal response uniformity, and temporal characteristics of the scintillation signals in the BGO crystals.

The ultimate goal is to establish a reproducible testing protocol for future characterization of calorimeter-grade crystals.

References

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